Product Requirements Document (version 1)

- **Objective**
  - **Goals**
    - A software that accurately predicts house pricing with complete GUI interface.
  - **Persona(s)**
    - To: Apartmentos 4 U Ltd.
    - From: R-AI
      - Product Owner: Raneem Abdel Hafez
      - Product Development: Raneem Abdel Hafez
      - Product Design: Raneem Abdel Hafez
- **Features & User Stories (add as many of these as required)**
  - Feature #1: Data Upload and Model Training
    - Description: Users can upload a CSV file containing house data (features and prices) to train a machine learning model.
    - User story: As a user, I want to be able to upload a CSV file of house data so that I can train a model to predict house prices based on the provided features.
    - Purpose: This feature enables users to quickly train a model on their data without needing extensive machine learning expertise.
    - User problem: Users need an easy way to train a machine learning model on their house data without having to write code.
    - User value: Users value the ability to tailor the model to their specific dataset, enabling more accurate price predictions for their target properties, and time-saving.
  - Feature #2: Price Prediction
    - Description: Users can input house features (bedrooms, bathrooms, house size, acre lot, zip code) and receive a predicted house price.

        - User Story: As a user, I want to enter the features of a house I'm interested in so that I can receive an estimated price before making a purchase decision.
        - Purpose: This feature empowers users to quickly get an estimate of a house's potential price, helping them make informed decisions about house purchases.
        - User Problems: Users want a simple way to obtain estimated prices for houses they are interested in.
        - User Values: Users value the convenience of obtaining estimated prices for properties they are interested in, enabling more informed negotiations, and buying decisions.
- **Technical Details**
    - Technologies to be used:
    - Backend: (Python)
        - Flask: A micro web framework for the backend to handle routing, requests, and responses.
        - Pandas: A data manipulation library to handle the processing of uploaded CSV files.
        - Scikit-learn: A machine learning library for training and evaluating the Random Forest Regressor model.
        - Joblib: A library for saving and loading trained machine learning models.
    - Frontend:
        - HTML: To structure the content and layout of the web pages.
        - JavaScript: To add interactivity and dynamic behavior to the frontend, GUI.
        - Fetch API: To make asynchronous requests to the backend endpoints.
        - CSS: For styling and presentation of the user interface.
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- **User Flow and Design**
  - Wireframes:
    - Home Page Wireframes
      - Title: House Price Estimation
      - Description: A brief description of the app's purpose.
      - Upload Data Button
      - Predict House Prices Button
      - Calculate Model Accuracy Button
    - Upload Data Page Wireframe:
      - File Upload Form:
      - Label: Upload CSV File
      - Input: File Input Field
      - Button: Upload & Train
      - Training Message: Displayed after successful model training.
    - Predict House Prices Page Wireframe:
      - Input Fields:
        - Label: Bedrooms
        - Label: Bathrooms
        - Label: House Size
        - Label: Acre Lot
        - Label: Zip Code
    - Predict Price Button
      - Predicted Price Display
    - Calculate Model Accuracy Page Wireframe:
      - Accuracy Calculation Display

    - Mockups: Posted as png
- **Analytics**
    - collected analytics from the app:
        - Accuracy of the product
        - Number of visitors/ Users
        - App performance
        - Accurate User data